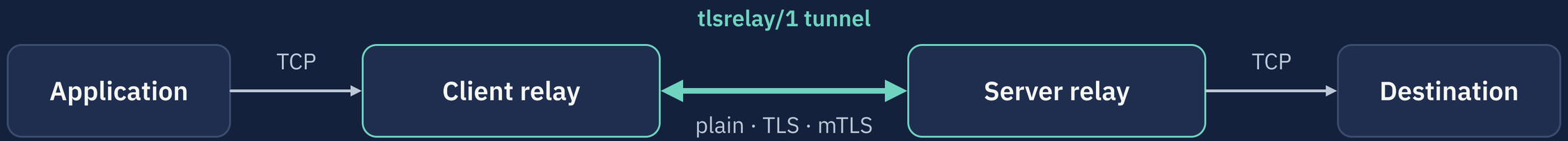
SOLUTION DESIGN · DELIVERIES 1.0 AND 1.1

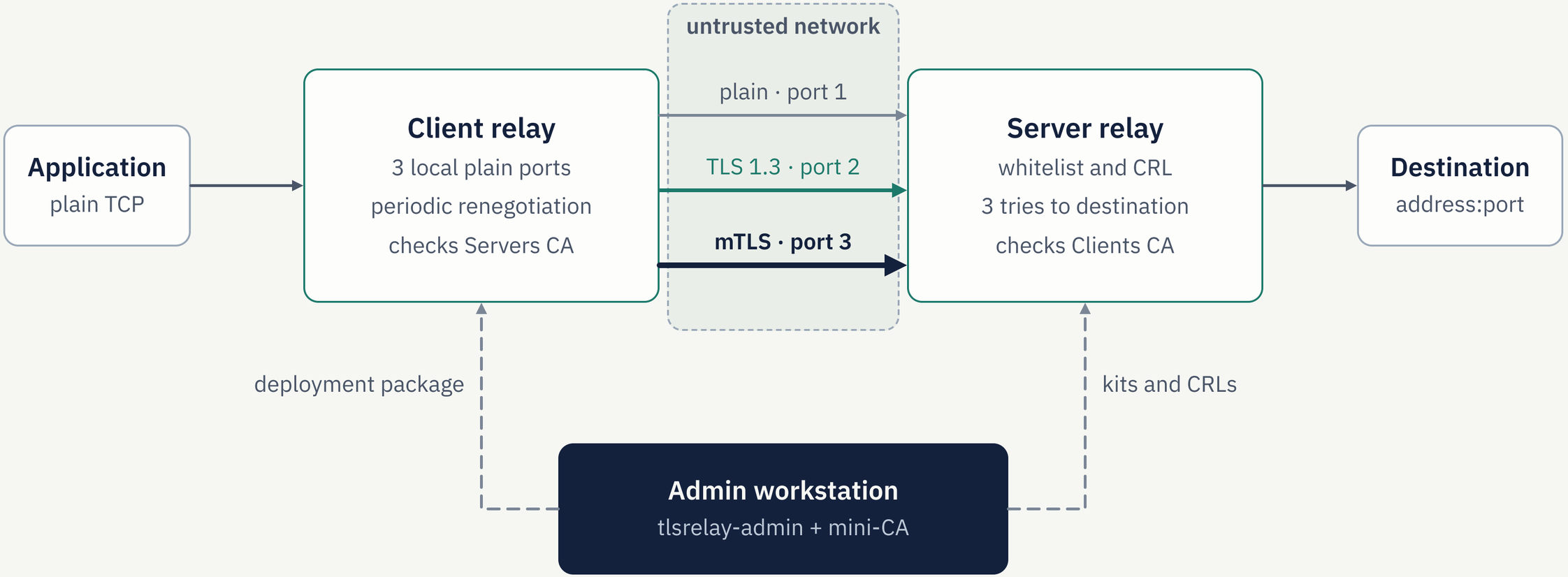
tlsrelay

A TCP relay that carries applications across untrusted networks in clear, over TLS 1.3 or mutual TLS, with transparent renegotiation and hot PKI rotation.



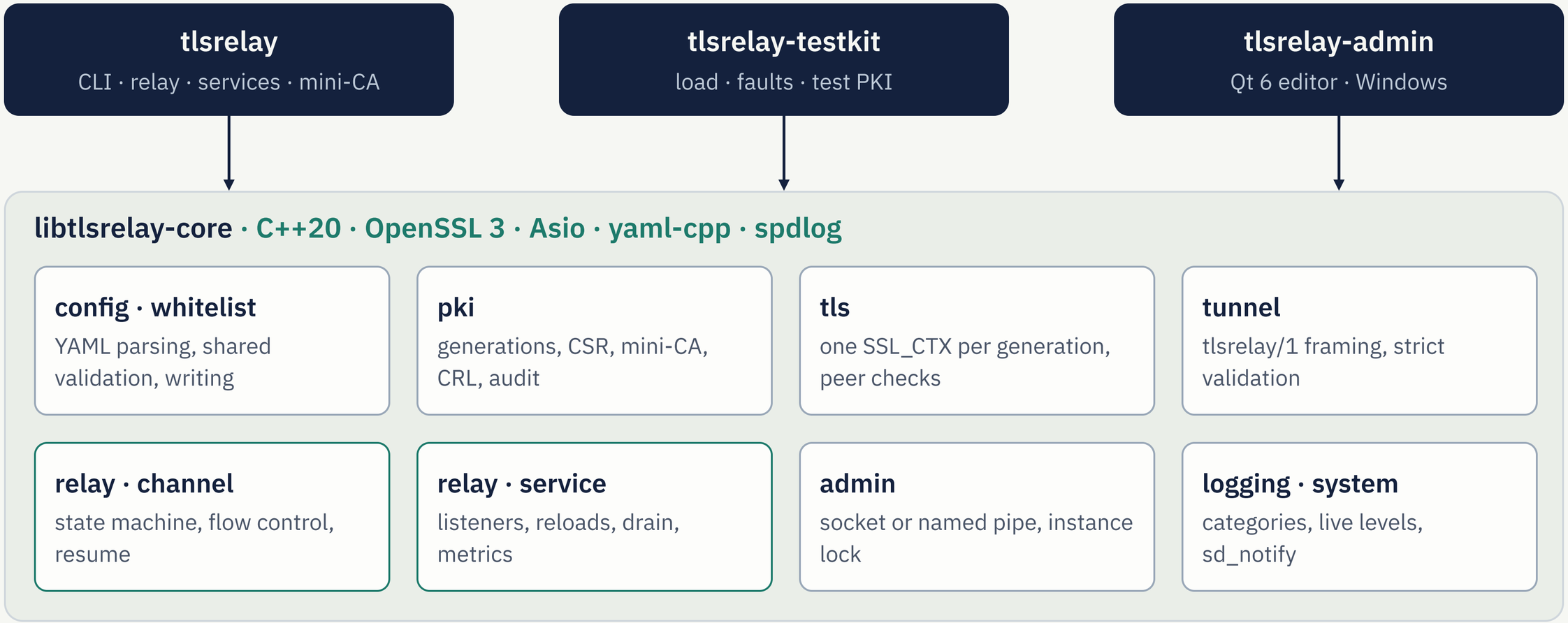
1 · SYSTEM CONTEXT

Two relays, three protection levels, one destination



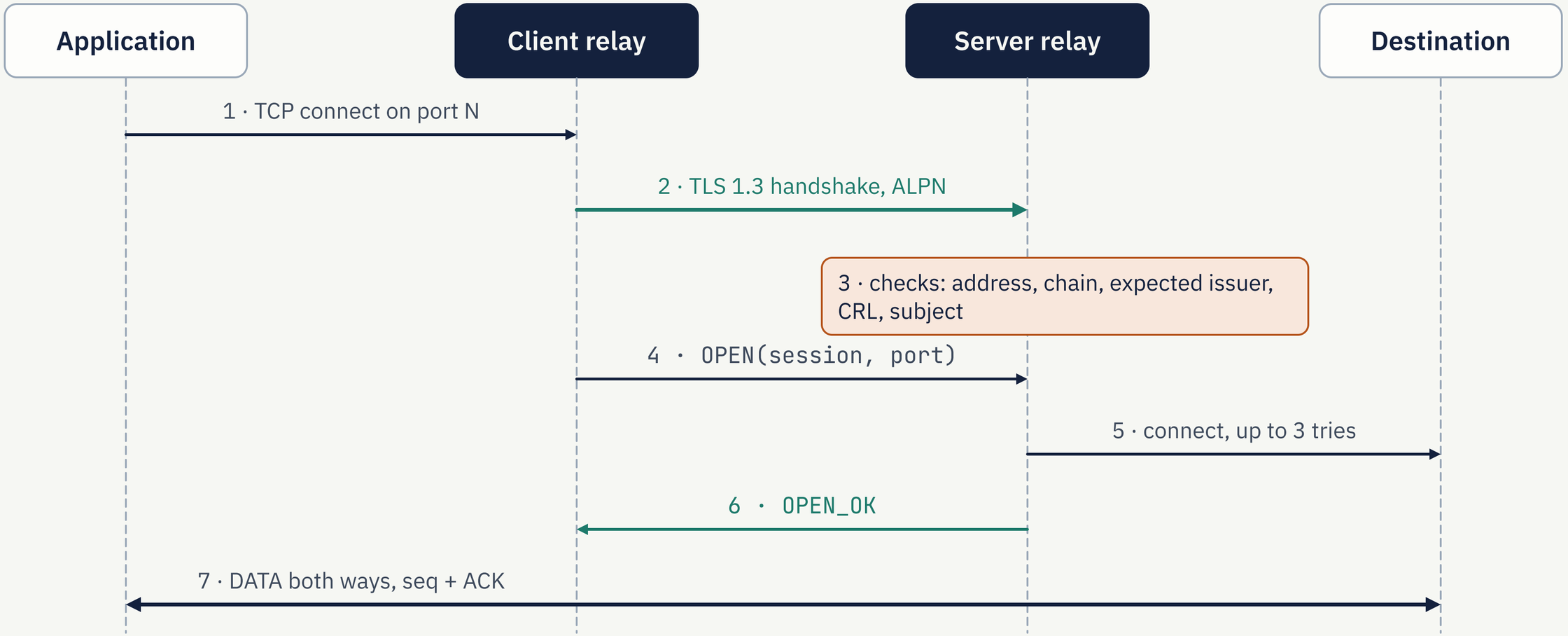
2 · ARCHITECTURE

One core library, three thin executables



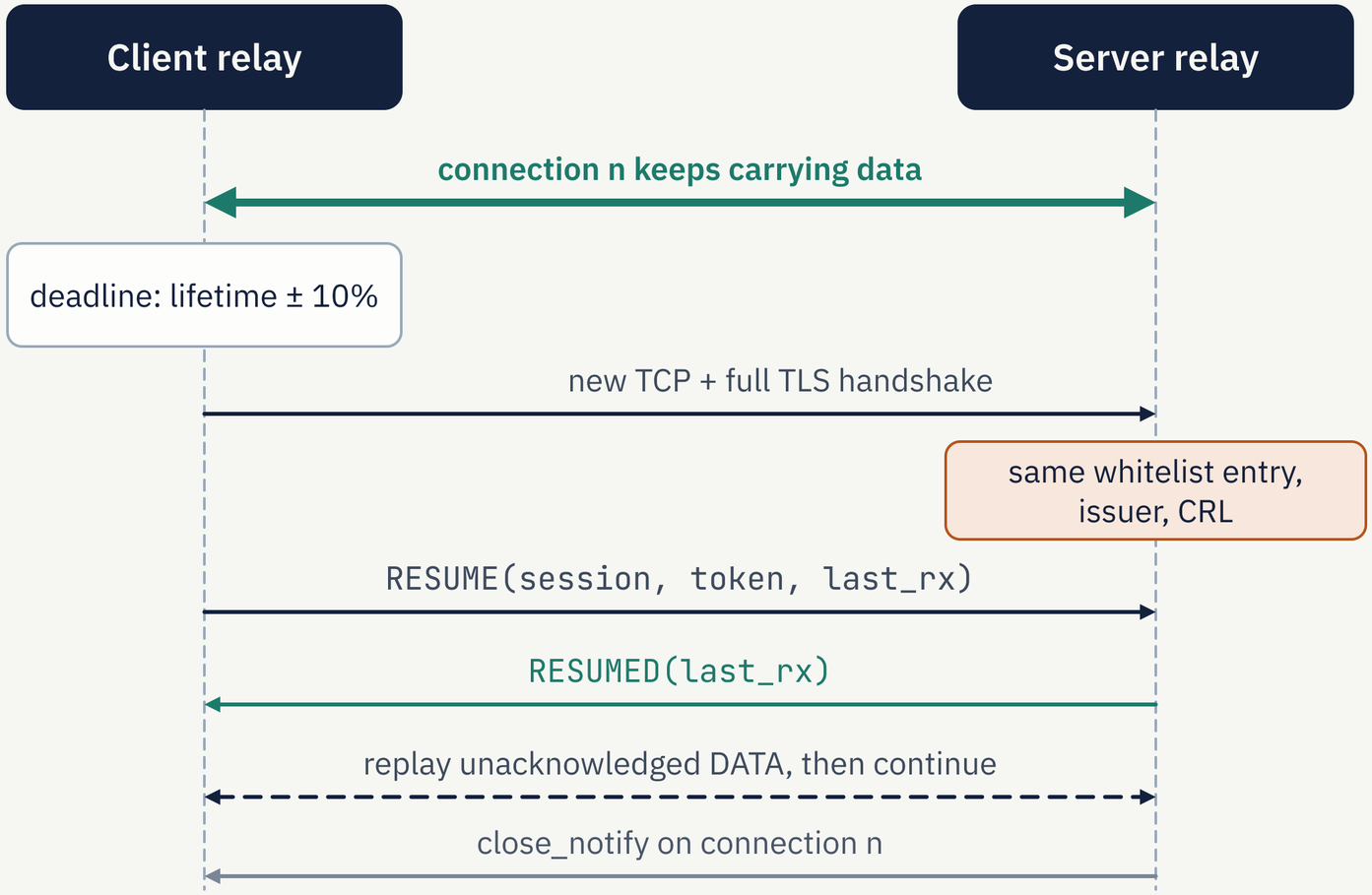
3 · OPENING A CHANNEL

All checks first, destination last



4 · TRANSPARENT RENEGOTIATION

A fresh TLS connection, without a pause



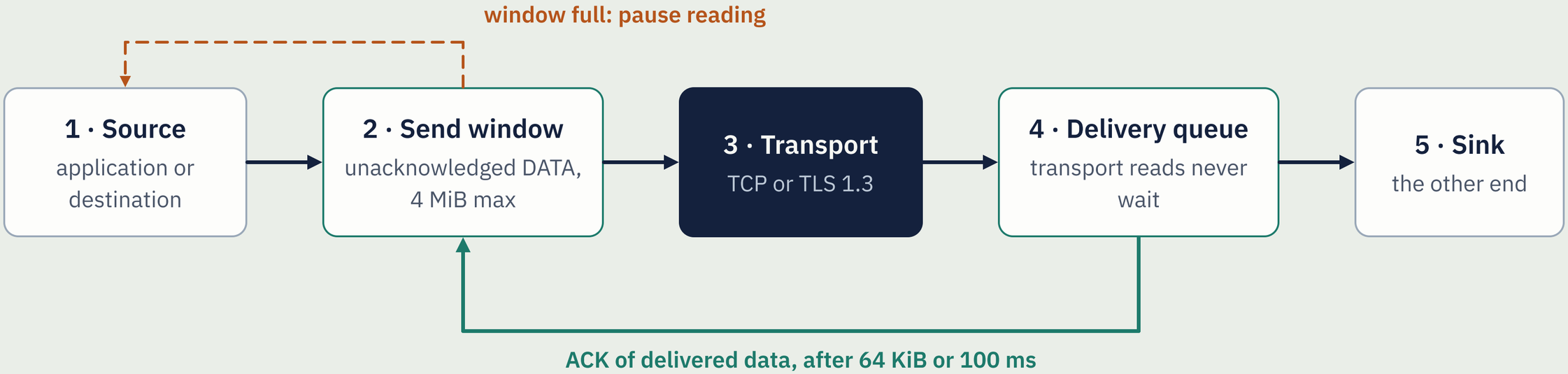
Resume token

```
token_n = TLS exporter(
"EXPORTER-tlsrelay-resume",
session_id || n, 32 bytes)
```

- Bound to connection n, used once
- Compared in constant time
- Retried until renegotiation_grace
- Same path resumes after a cut
- Plain port: no resume, idle only

5 · DATA PATH AND FLOW CONTROL

Bounded buffers: back-pressure instead of loss



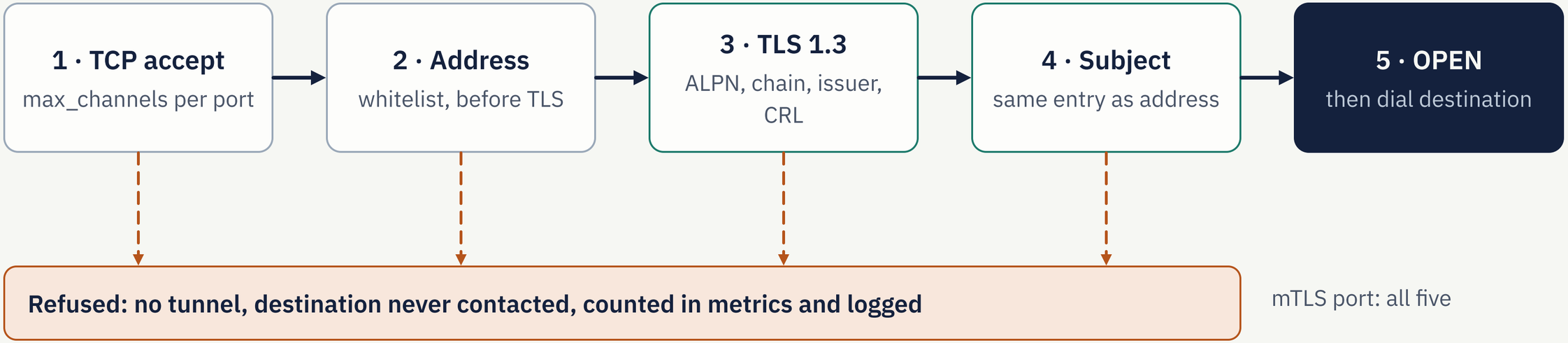
4 MiB
resume buffer per direction and per channel

256 KiB
coalesced per write, in both directions

15 s × 3
PING keepalive: a silent link is dead after 45 s

6 · SECURITY

Five gates before a server relay opens a tunnel



Keys

Instance keys never leave their host. CA keys: scrypt + AES-256, never stored by the editor.

Hardening

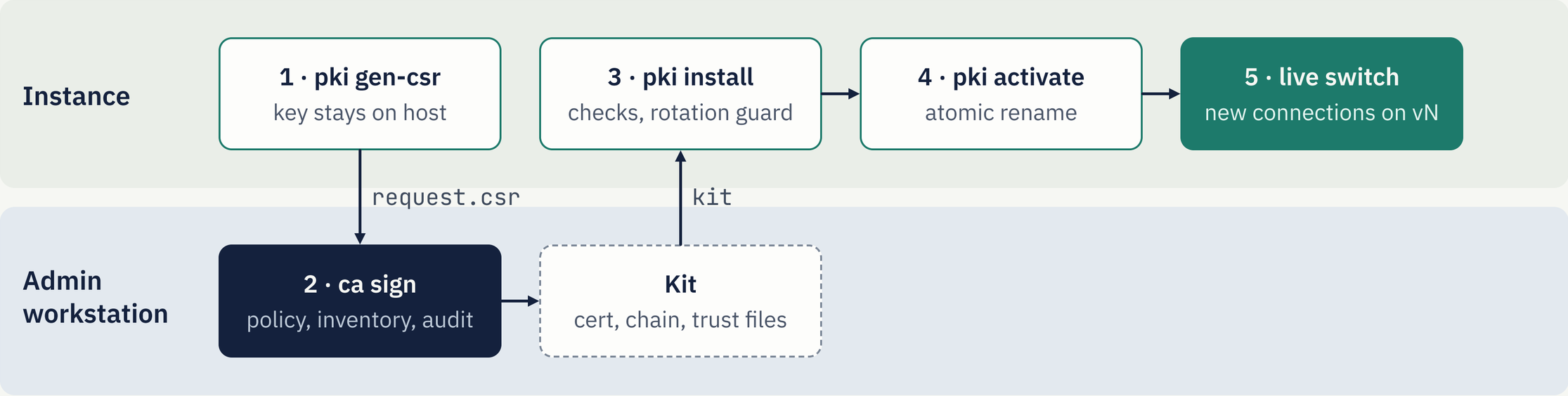
TLS 1.3 only, no tickets. Sandboxed services. Sanitizers, fuzzing, static analysis in CI.

Operations

Local admin channel only. No secret or relayed data in logs; peer text escaped.

7 · PKI LIFECYCLE

Keys stay home; generations switch atomically



Generations

v1 · draining → v2 · current

Open channels move to v2 at their next renegotiation; v1 is freed when unused.

Revocation

`ca revoke → CRL → pki crl`

Revoked peers are refused at the handshake and their open channels are closed.

8 · ROOT CHANGE

Three steps, every instance, no cut

Each step reaches every instance before the next one starts



Trust roots	old + new	old + new	new only
Leaf certificate	renewed by old CA	signed by new CA	renewed by new CA
Expected issuers	old + new	old + new	new only

Prepared and verified by the tooling

- tlsrelay ca root-change, or the editor wizard, writes the step 1 and 2 trust files per role, the combined CRLs and the exact commands
- The rotation guard accepts each step and refuses any shortcut to the new root
- Acceptance scenario R09 runs all three steps under continuous traffic and checks data integrity

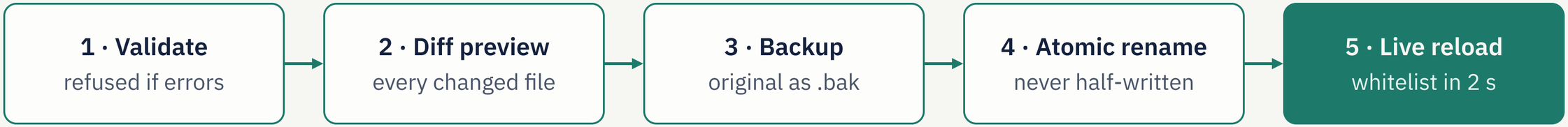
9 · EDITOR AND DELIVERY

tlsrelay-admin reuses the service's own rules

VALIDATION · the typed text goes through the service loader



SAVE · nothing is written until it is valid and reviewed



196
tests: GoogleTest, Qt Test, lifecycle scripts

18
CI jobs on every push, Linux and Windows

5
Linux distributions where packages are installed in CI

5,000
concurrent channels, about 800 Mbit/s each